

TECHATHON 1.0

IUT
ROBOTICS
SOCIETY



SMART
FIREFIGHTER

Jilapi

SITUATION ANALYSIS

MARKET INSIGHT FOR DRONES:

- **Main Purpose:** Spread drone over mass community of people.
- **Environmental Monitoring:** Monitoring environment, wildlife, vegetation and forest, assessing air and water quality, aiding precision agriculture and supporting disaster response.
- **Technological Advancements:** Drone technology advances, like longer battery life and obstacle avoidance, are fueling market growth by expanding application.
- **Global Market:** Emerging economics where both commercial and govt. sectors are adopting drones, offering growth among industry players.

STATUS:

- Provide B2B and B2G service

OBJECTIVES

- Enhancing technical specifications.
- Sustainability and versatility.
- Marketing and monetization.
- Financial perspective.



PROBLEM IDENTIFICATION

Core ISSUE

1

As of late there are numerous huge fire mishap occurred in Bangladesh

2

Many individuals as well as many fireman licked the bucket

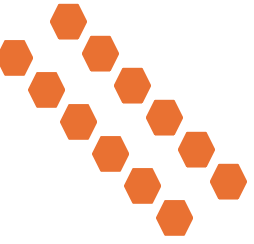
3

The primary explanations for the passings are:- Over the top intensity, smoke and in the particular sluggish strategy of the putting out fires framework

THE GRAPHICAL STRUCTURE
ALSO SHOW THE **RAPID
INCREASING LEVEL OF FIRE
MISHAP** IN THE RECENT
YEARS IN DHAKA AND
BANGLADESH

TARGET GROUP ANALYSIS

PRIORITY WISE HIGHER ORDER SEGMENTATION



Fire Departments

- Are primary buyers of drones for aerial surveillance
- Aid in situational awareness
- Assists in resource deployment during fire fighting operation



Emergency Response Agencies

- Invest in drones
- Enhance capabilities in assessing disaster scenes
- Aid in coordinating rescue efforts
- Are used by search and rescue organization



Govt. Entities

- Government bodies procure drones for fire safety and disaster management
- Bolster firefighting and emergency response infrastructure
- Are used at local, regional and national levels
- Enhance the effectiveness of government-led firefighting and emergency response effort



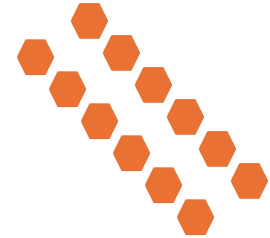
Industrial Facilities

- Large industrial facilities purchase drones for fire prevention.
- Enhance firefighting capabilities
- They are utilized in high-risk environments
- Aid in on-site fire prevention and response at factories, refineries and powerplants

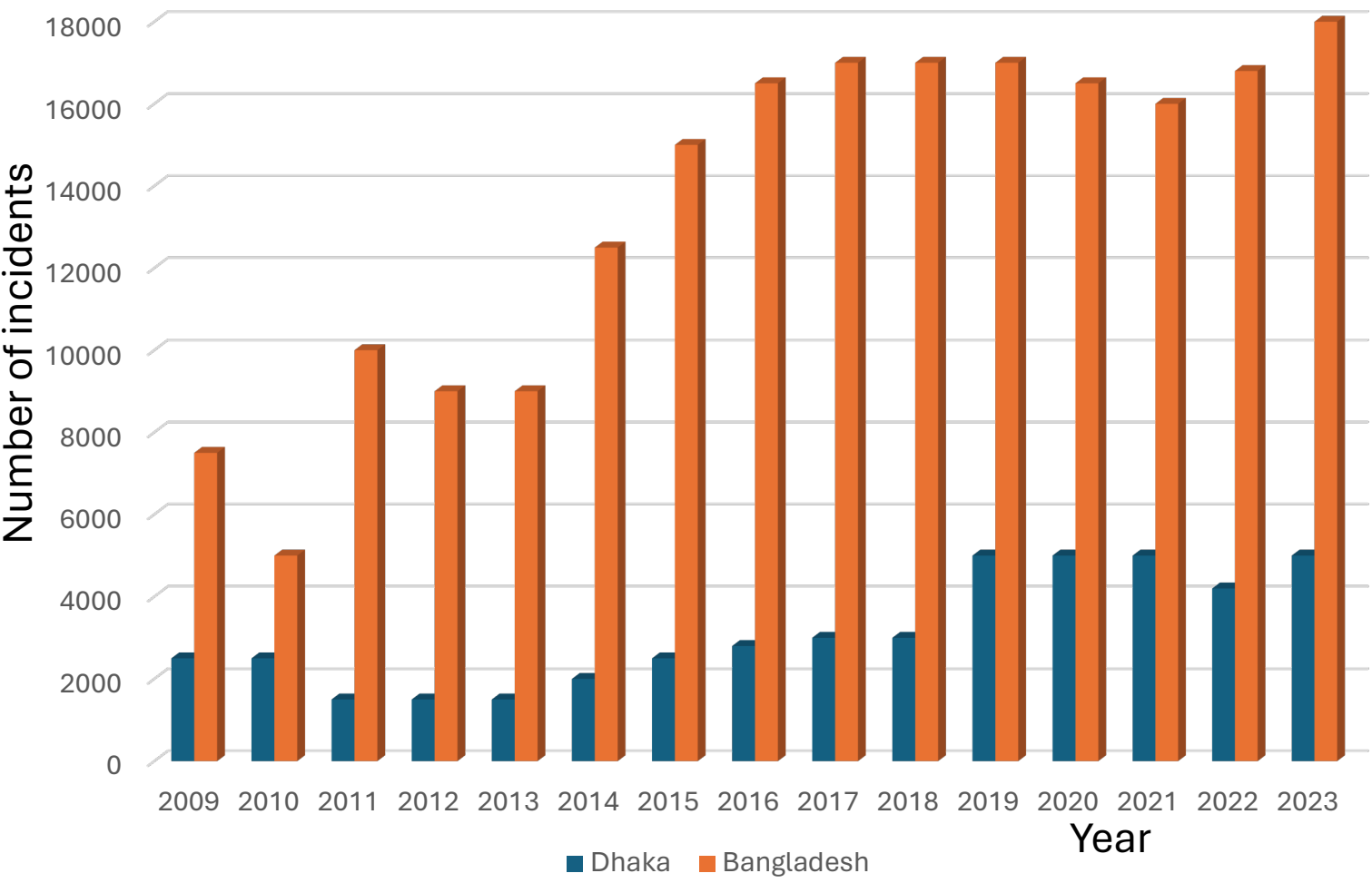


Private Conductors

- Offering customized drone solutions
- Specialize in firefighting services, aerial surveillance or drone operations
- Private companies may target fire service agencies as clients
- Solutions are tailored to specific firefighting needs and requirements



Year-wise number of fire incidents



SEGMENTATION



GEOGRAPHY

- Urban
- Sub-Urban
- Rural

INSIGHT:

Fire department from urban and sub-urban areas are the primary TG and with highest potential

BIG IDEA



As we are unable to produce a firefighting drone in our own industry due to lack of resources, we have to think a different way.

SKYF, The firefighting drone producer, is very well-known all over the world. So we have some proposal to modify this drone at this aspect:-

- ☐ Firstly we will give the proposal to “**SKYF**” company for hiring their retired scientists and recruit in our own industry.
- ☐ Then we will make a proper budget for buying the essential necessary things from the others countries.
- ☐ Then we will upgrade our company into a next step, will train the workers and expertizers and then produce firefighting drone

We will also bring some changes in modification for Agricultural, real state photography and delivery service purpose
Like:-

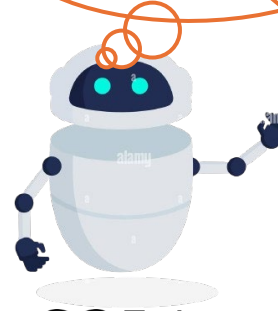
1. Integrate multispectral or hyperspectral cameras for advanced crop monitoring and analysis
2. Integrate smart flight modes and obstacle avoidance sensors for safe and precise aerial maneuvers around buildings
3. Integrate advanced GPS and navigation system for precise positioning, especially in complex urban environments.



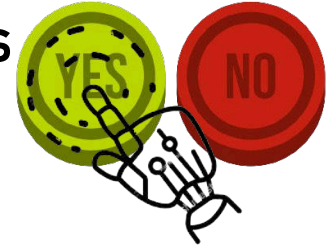
ACTION PLAN

- ✧ To respond effectively to building fire, we suggest **employing a two-wheeled vehicle** for better **maneuverability and ample space to carry** robots and equipment.
- ✧ The robots will feature **vespel piping** to withstand high temperature (**up to 3000°C**), enabling them to operate **efficiently in extreme heat conditions**.
- ✧ Employing drones like the **SKYF Robot**, renowned for its **high payload capacity** and **aerodynamic performance**, to aid in firefighting operations
- ✧ Depending on the filler material (**unfilled, 15% graphite, 10% PTFE and 15% Moly**), vespel can withstand **350 hours of 398 degree Celsius heat**, losing only 50% of its initial tensile strength. It can take up to **12500 psi**.

Presently there can be an inquiry: emerge that might a robot, at any point, do a line loaded with water??



The Response is



- ✧ Enhancing the robots durability in extreme temperatures by **integrating materials** like **tantalum carbide** and **hafnium carbide** (endure up to **4000°C**)
- ✧ Deploying multiple specialized robots equipped with functions such as **water delivery, carbon dioxide foam application, smoke extraction and medical assistance** to effectively combat fires.
- ✧ Implementing a **forklift system** to supply water and materials to the robots, addressing **gravitational challenges** during high-rise emergencies,
- ✧ Outfitting each robot with cameras for **real-time monitoring** and **strategic firefighting maneuvers**, including **fire location identification** and **victim rescue**.

Contents	SKYF Drone	DJI mini 2
		
Weight	250kg (lowest) (lbs)	400 kg (highest)- 249 grams
Camera	N/A	12 megapixel
Flight Reach	Upto 350 km	Upto 10km
Size	5.2x 2.2m ²	.138x 0.081x 0.058 m ³
Battery Life	1200 mAh	2250 mAh
Flight Duration	Upto 8 hours	Upto 31 minutes
Maximum speed	70 km/h	57.6 km/h
Position Accuracy	Within 30cm preciton	N/A
Max Payload	400kg	N/A

ACTION PLAN



Bringing Firefighting to New Heights

|Your Aerial Guardian

Tagline



WEB BASED MARKETING

- **Website Excellence:**

Create an optimized website to highlight firefighting drone features and benefits

- **Social Engagement:**

Utilize social media and email marketing to share captivating content and promotions, engaging with potential customers and building relationships

- **Expanded Reach**

Employ online advertising, influencer partnerships and webinars to extend brand reach and educate audiences on the value of firefighting drones.



E-COMMERCE EXPANSION

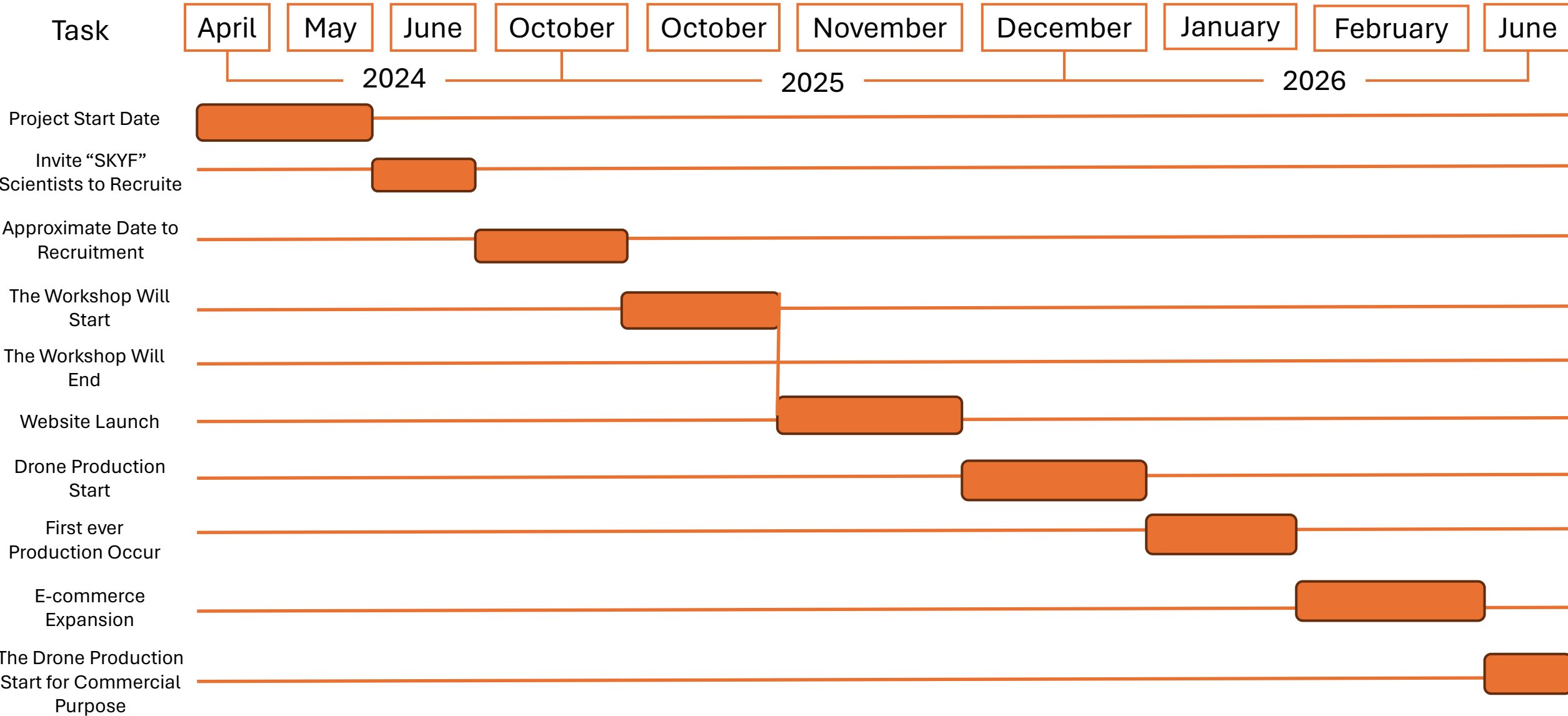
- **Targeted Digital Marketing**

Implement focused digital marketing campaigns, including SEO, content marketing and paid advertising, to reach and engage potential customers interested in firefighting equipment.

- **Global Outreach:**

Explore international expansion opportunities, collaborate with industry partners and provide localized support.

TIMELINE & ROADMAP



COMPETETIVE MARKET ANALYSIS



The global firefighting drone market was valued at **\$0.97 billion in 2021**, and is projected to reach **\$2.4 billion by 2031**, growing at a CAGR of **10.2% from 2022 to 2031**.



FIREFIGHTING DRONE MARKET

OPPORTUNITIES AND FORECAST, 2021 - 2031

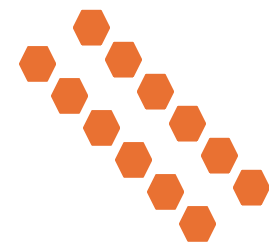
Firefighting drone market is expected to reach **\$2.4 BILLION** by 2031

Growing at a **CAGR OF 10.2%** (2022-2031)

Report Code: A06280. www.alliedmarketresearch.com

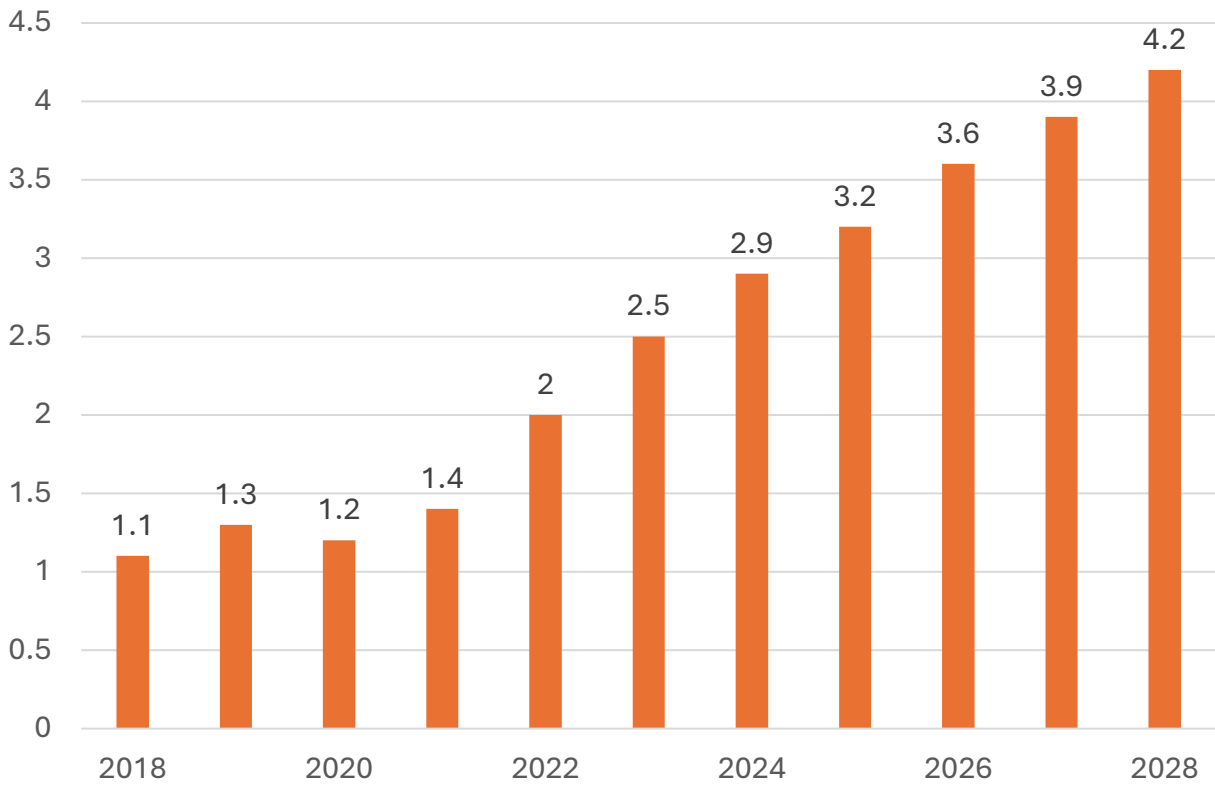
Firefighting Drone Market Report Highlights		
Aspects	Details	
 Market Size By 2031	USD 2.4 billion	
 Growth Rate	CAGR of 10.2%	
 Forecast period	2021 - 2031	
 Report Pages	278	
 By Type	- Fixed Wing - Multi-rotor	
 By size	- Micro Drones - Macro Drones	
 By Propulsion	- Electric Motor - Piston Engine - Solar Powered	
 By Application	- Scene Monitoring - Search and Rescue - Post Fire or Disaster Assessment - Firefighting	

COMPETETIVE MARKET ANALYSIS

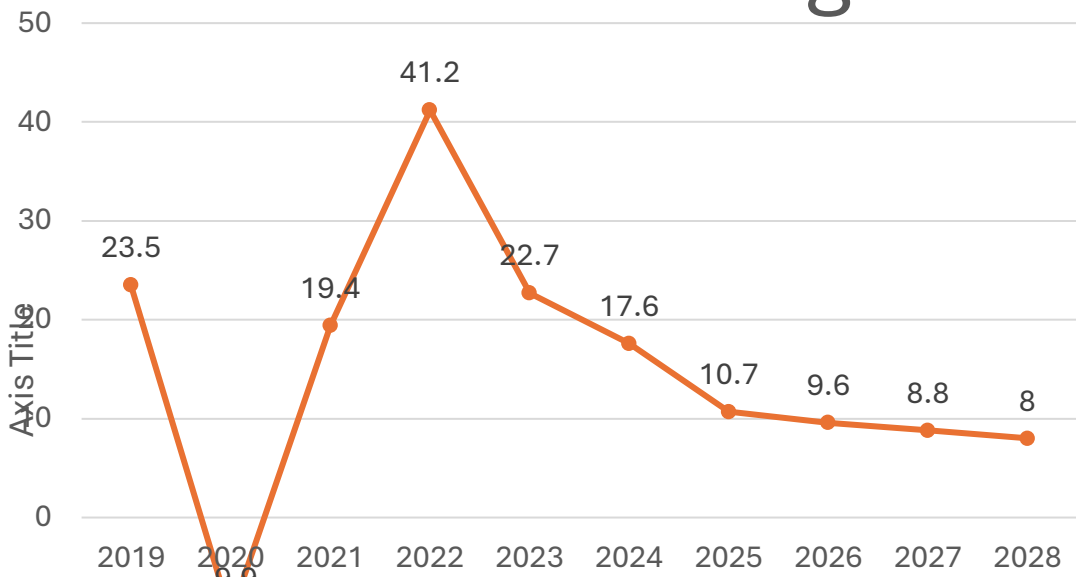


In million USD

Revenue



Revenue Change



Most recent update: Mar 2024
Source: statista Market Insights

FINANCIAL FEASIBILITY ANALYSIS

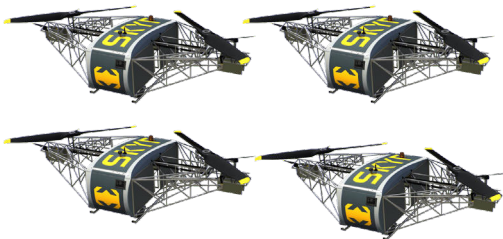
Material/Events	Cost/Unit (BDT)	Unit	Month	Total Cost
Vespel Line	30,000 per kg	1		30,000
Hafnium Carbide	20,000 per kg	1		20,000
Web-based Marketing	1 Lac	1	3	3 Lac
E-commerce Expansion	5 Lac	1	2	10 Lac
			Grand Total=	13,50,000/=



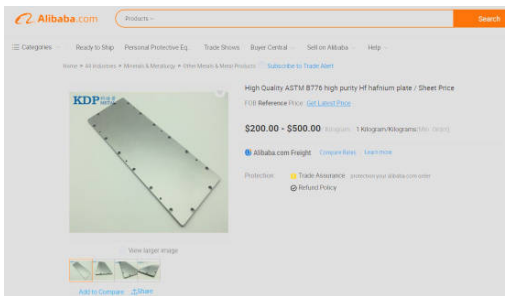
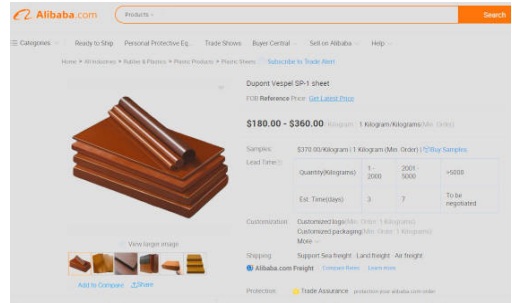
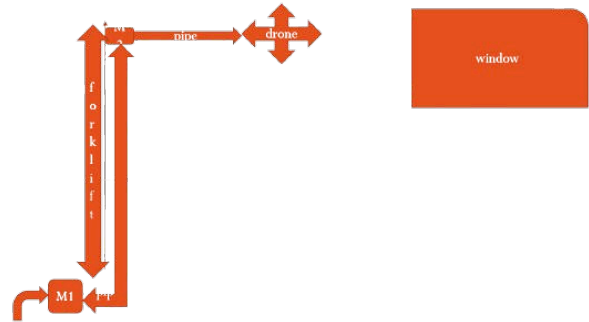
Price =
80 Lac – 1 Crore



Price =
10 Lac – 15 Lac



Close to
half cost of
one fire
administration
vehicle



MONITORING & EVALUATION



**Website Publication and
Release It Any Social Platform**



**Contract Another Similar
Company Like “Copter Express”**



Performance Metrics



Analysis And Reporting



**Survey
Results**



Feedback Mechanism

CONTINGENCY PLAN

- ❑ If we fail to meet our desire goal, we will then have to move on to subsidiary plan. If the invited scientists will decline to join with us, we will give this proposal to the **Russian drone company, "SKYF"** directly with **G2G** or **B2B** service. Then we will order them to produce firefighting drone. In that case, we will deliver them **mini DJI 2** drone and then they will convert it into the firefighting drone. After receiving and applying it in several fire mishaps in our country. And this will save thousands of lives every year where increasing cost is not a big deal.

ROADMAP



Tech spot



SKYF



Drone Modifying



Get Firefighting Drone



Save Thousands of live